

Akshat Srivastava

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SKILLS

- **Programming Languages:** Embedded C, C, Python, Assembly, Java
- **Microcontrollers & Boards:** ESP32, Arduino, Raspberry Pi
- **Embedded & IoT:** Sensor Interfacing, Real-Time Monitoring, IoT Architecture
- **Communication Protocols:** UART, SPI, I2C, GSM, GPS, nRF24L01
- **EDA & Simulation:** MATLAB, Simulink, Verilog, Keil µVision
- **Design Tools:** AutoCAD, Creo, KiCad
- **Soft Skills:** Problem Solving, Analytical Thinking, Team Collaboration, Willingness to Learn, Cross-functional Communication.

INTERNSHIP

Super Tannery Ltd. – Process & Operations Intern

Jun 2025 – Jul 2025

- Analyzed end-to-end manufacturing workflows across 5+ production stages in a large-scale shoe manufacturing unit.
- Monitored production efficiency for a line producing 1,000+ units per day.
- Supported quality control inspections, helping identify defects and reduce rework during daily checks.
- Observed industrial automation and process optimization practices used in high-volume manufacturing.

PROJECTS

Autonomous IoT Boat for Waterborne Disease Detection (SIH 2025) ([Link](#))

Sep 2025

- Designed an autonomous IoT-based embedded system using ESP32 and water-quality biosensors.
- Implemented GPS and ultrasonic sensors to enable obstacle detection and autonomous navigation.
- Streamed 100+ real-time sensor readings/day via GSM to a centralized web portal during flood scenarios..
- Improved operational safety by enabling 24/7 remote monitoring without human intervention.

Technologies: ESP32, IoT, GPS, GSM

Remote Control for Actuators ([Link](#))

Mar 2025

- Architected a wireless actuator control system using Arduino Nano and nRF24L01.
- Deployed reliable wireless communication over long-range distances in interference-prone environments.
- Reduced system cost by over 60%, lowering unit cost from ₹5,000+ to under ₹2,000.
- Optimized hardware debugging and protocol tuning to ensure consistent actuator response.

Technologies: Arduino Nano, nRF24L01, Embedded C, Wireless Communication

Smart Energy Meter (IoT-Based System) ([Link](#))

Dec 2023

- Engineered an IoT-enabled smart energy meter using ESP32 and PZEM-004T for real-time power monitoring.
- Integrated continuous monitoring of voltage, current, power, and energy with sampling at 2–3 second intervals.
- Developed a web dashboard displaying live electrical parameters with 2–3 second refresh intervals for 24/7 monitoring.
- Enabled threshold-based alerts to notify users when energy usage exceeds predefined limits.

Technologies: ESP32, Embedded C, Sensors, WiFi, IoT, Web Server

CERTIFICATES

- TCS iON Career Edge – Young Professional Aug 2025
- AI for OCI and Cloud Applications – Oracle Jan 2025
- Embedded C Programming for Microcontrollers – Udemy Aug 2024
- Python Programming – Scaler Dec 2023

ACHIEVEMENTS

- 2nd position, Robo Offroad Racing – Utkansh, NIT Jalandhar April 2025
- 2nd position, Robo Soccer - Xpecto, IIT Mandi Mar 2025
- 3rd position, Robo Sumo - Cognitia Feb 2025
- 1st position, Tech Fest - Pushp Gujaral Science City, Jalandhar Sep 2024
- 1st position, Circuit Solving - Tech Marathon (SEEE). Dec 2023

EDUCATION

Lovely Professional University Bachelor of Technology Electronics and Communication Engineering; CGPA: 6.84	Phagwara, Punjab Aug 2023 - Present
K.R. Education Centre Intermediate 75%	Kanpur, Uttar Pradesh May 2022 - Mar 2023
K.R. Education Centre Matriculation 83%	Kanpur, Uttar Pradesh Apr 2020 - Mar 2021